PROGRAMMING IN JAVA

Assignment 06

TYPE OF QUESTION: MCQ

Number of questions: 10 Total marks**: 10 × 1 = 10**

QUESTION 1:

Which method is used to start a new thread in Java?

a. `run()`

b. `start()`

c. `execute()`

d. `begin()`

Correct Answer: b

b. start()

Detailed Solution:

Calling `start()` creates a new thread and internally invokes the `run()` method. Calling `run()` directly does not create a new thread.

QUESTION 2:

Which interface must be implemented to create a thread using the Runnable approach?

a. `Threadable`

b. `Runnable`

c. `Callable`

d. `Executor`

Correct Answer: b

b. Runnable

Detailed Solution:

The `Runnable` interface contains a single method `run()` and is used when a class needs to be executed by a thread.

QUESTION 3:

What happens if a thread's `run()` method completes execution?

a. **The thread enters the waiting state**

b. **The thread can be restarted**

c. **The thread enters the terminated state**

d. **The thread returns to the runnable state**

Correct Answer: c

c. **The thread enters the terminated state**

Detailed Solution:

Once the `run()` method finishes, the thread completes its lifecycle and enters the terminated state. A terminated thread cannot be restarted.

QUESTION 4:

Which keyword is used to synchronize a method in Java?

a. `Lock`

b. `Threadsafe`

c. `Synchronized`

d. `Volatile`

Correct Answer: c

c. Synchronized

Detailed Solution:

The `synchronized` keyword ensures that only one thread can access a method or block at a time, preventing race conditions.

QUESTION 5:

Which of the following is the default priority of a Java thread?

a. **1**

b. **3**

c. **5**

d. **2**

Correct Answer: c

c. **5**

Detailed Solution:

Java thread priorities range from 1 (`MIN_PRIORITY`) to 10 (`MAX_PRIORITY`). The default priority is 5 (`NORM_PRIORITY`).

QUESTION 6:

Which method causes the currently executing thread to pause for a specified time?

a. `wait()`

b. `sleep()`

c. `pause()`

d. `delay()`

Correct Answer: b

b. sleep()

Detailed Solution:

The `sleep()` method pauses the currently executing thread for a specified number of milliseconds without releasing any locks.

QUESTION 7:

Which exception is thrown when `sleep()` is interrupted?

a. **RuntimeException**

b. **IOException**

c. **InterruptedException**

d. **ThreadException**

Correct Answer: c

c. **InterruptedException**

Detailed Solution:

`sleep()` throws `InterruptedException` if another thread interrupts the sleeping thread.

QUESTION 8:

Which class is used to read text data efficiently from a file in Java?

a. `FileInputStream`

b. `BufferedReader`

c. `DataInputStream`

d. `ObjectInputStream`

Correct Answer: b

b. BufferedReader

Detailed Solution:

`BufferedReader` reads text efficiently using buffering and provides the `readLine()` method for line-by-line input.

QUESTION 9:

Which method is used to read a line of text using `BufferedReader`?

a. `read()`

b. `readText()`

c. `readLine()`

d. `getLine()`

Correct Answer:

c. readLine()

Detailed Solution:

The `readLine()` method reads an entire line of text and returns it as a `String`. It returns `null` at end of file.

QUESTION 10:

Which class is the superclass of all input byte streams in Java?

a. **Reader**

b. **InputStream**

c. **FileInputStream**

d. **BufferedInputStream**

Correct Answer:

b. **InputStream**

Detailed Solution:

`InputStream` is the abstract superclass for all byte-based input streams in Java, such as `FileInputStream`.